High gain fusion in a Staged Z-pinch

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***Abstract*—The implosion of a Staged Z-pinch is simulated for parameters that are characteristic of the ZR accelerator, at Sandia National Laboratories. The Staged Z-pinch configuration is comprised of a cylindrical, Silver (Ag) plasma shell, 3-mm outer radius, 0.01-cm thick, imploding onto a uniform-mixture fill (target) of Deuterium-Tritium (DT); the Z-R parameters are: 130 ns, 27 MA, 22 MJ; and the simulation code is MACH2, a 2-1/2 D, radiation-MHD code. During implosion magnetosonic shock waves are generated which propagate radially inward at different speeds in the liner and target plasmas, producing a shock front at the interface between the liner and target plasmas, and a conduction channel ahead of the liner that facilitates target preheating. The Ag/DT interface remains stable throughout the compression, even as the outer surface of the liner becomes RT unstable. At peak compression a string of hot spots forms in the target plasma, that triggers ignition and a fusion-energy yield in excess of 200 MJ; net-energy gain approaching a factor of 10, based on the total stored energy.**

***Index Terms*——Z-pinches, Rayleigh Taylor Instability, magneto-inertial fusion, shock waves**

I. INTRODUCTION

The Staged Z-pinch is a magneto-inertial fusion concept comprised of a thin, high-atomic-number liner imploding onto a low-atomic-number (fusion) target [1], [2]. Studies in the 50 GW regime ($\geq$ MA, μs) provide experimental evidence for staging [3]; the concept is scalable to the 100 TW regime ($\geq$ 15 MA, 0.1 μs). The role of shock waves for stable implosion and target heating plays crucial role in the production of high energy density plasma [4]. High-yield fusion has been predicted for a gas load pinch with a diameter of order, $r \sim$ cm (needed to achieve high implosion velocity [5], [6]) and for the previous parameters of the Sandia Z-facility. There is new interest in configurations where a solid liner implodes onto an embedded, axial-magnetic field, and target that is preheated by a laser [7], [8], [9].

Conventional Z pinches are susceptible to a variety of instabilities, which limit the attainment of extreme-energy density. The RT instability [10], [11] has the fastest growth time. Techniques have been used to mitigate RT instabilities, with varying degrees of success. Previous Staged Z-pinch experiments [3] suggest that an axial magnetic field will help, but will inhibit a tight implosion and production of high energy density plasma.

The Staged Z-pinch mitigates the RT-instability by choosing a small-initial radius, since the growth rate is proportional to the distance over which the liner accelerates. When imploded, magnetosonic shock waves propagate inward at different speeds in the liner and target plasmas. A direct benefit of the shock waves is formation of a conduction layer at the interface, that remains stable and carries the main-current pulse at peak implosion. Target preheating occurs before the final implosion, when the liner becomes RT unstable. Flux leakage through the liner produces a large component of azimuthal-magnetic field between the liner plasma and the conduction layer near peak implosion due to flux compression; this field confines fusion α particles, and insulates the target plasma from the liner plasma.

The sections of this paper are organized as follows; Section II describes the results from the simulation, using the MACH2 code to model the approximate experimental configuration, and Section III concludes the paper.

II. SIMULATION

The simulation considers the implosion dynamics of the staged Z-pinch where a liner of silver plasma implodes onto a DT target plasma. The simulation uses the 2-1/2 D, MHD code, MACH2 [12], [13]; the code has a two-dimensional grid, but computes three components of the vector fields for velocity and magnetic field. MACH2 is a time-dependent, Arbitrary Lagrangian Eulerian (ALE) simulation code that solves for the continuity, momentum, energy, and magnetic field equations on a grid composed of quadrilateral cells in the r and z plane. The code uses a finite volume differencing technique and either analytic models, or tabular values (SESAME tables at Los Alamos National Laboratory), for the equation-of-state and transport variables. In modeling the high energy density staged Z-pinch, a simple radiation emission model employing non-LTE (non-local thermodynamic equilibrium) opacities was used. This radiation model works quite well during the implosion when the plasma is optically thin, but is not expected to give quantitative results for the radiation pulse at the peak implosion when plasma density is comparable to solid density.

The ZR generator [14] was modeled as an equivalent R-L circuit driven by the measured voltage waveform derived from reference [14]. For these simulations the series resistance was 0.18 ohm and the inductance 6.64 nH. Figure 1 shows the corresponding current waveform obtained from a typical MACH2 simulation run; it reproduces the experimentally and numerically obtained waveform in the reference.

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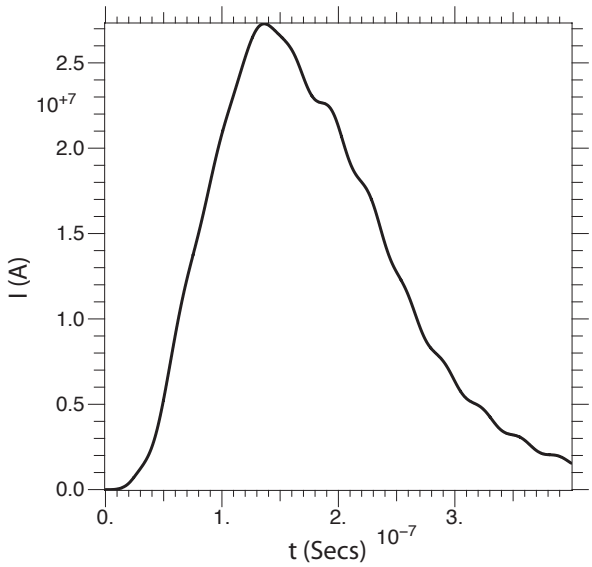

Fig. 1. (Simulated) Z-pinch current profile for the Sandia ZR Facility.

TABLE I
SIMULATIONS FOR THE ZR GENERATOR: AG LINER IMPLOSIONS ONTO DT TARGET.

Load configuration	Ag-DT
Initial pinch radius [cm]	0.3
Liner thickness [cm]	0.01
Liner mass density [kg/m^3]	800
Target mass density [kg/m^3]	9.8
Implosion Time, τ_i [ns]	126.4
Dwell Time, τ_d [ns]	1.0
Peak Ag-Ion Density, $\hat{n}_i$ [10^{25} cm^{-3}]	1.27
Peak H-Ion Density, $\hat{n}_i$ [10^{25} cm^{-3}]	1.3
Peak Ion Temperature, $\hat{T}_i$ [keV]	4
Peak Azimuthal Magnetic Field, $\hat{B}_\theta$ [MG]	527
Minimum Radius, R_f [cm]	0.008
Peak Kinetic Energy, $\hat{E}_k$ [MJ]	1.0
Ion Thermal Energy, E_i [MJ]	11.0
Total fusion Energy, E_f [MJ]	200.0

Our focus is to model a thin-silver liner, 6 mm diameter, that implodes on to a low-density, solid-fill DT plasma. Initial parameters used in the simulation and peak values obtained after the implosion are summarized in Table I. An initial density perturbation of one percent is applied throughout the simulation region. The simulations were run with 5 computational regions. All five regions extend the full length of the plasma in the axial direction (0.0 to 1.5 cm). The first region extends from the axis to 0.05 cm in the radial direction. The second region extends from 0.05 cm to 0.1 cm, the third from 0.1 cm to 0.2 cm, the fourth from 0.2 cm to 0.3 cm and the fifth region from 0.3 to 0.31cm. All five regions were discretized into 128 zones axially. The first, third and fourth regions were discretized into 32 zones each radially, the second region was discretized into 16 zones radially, and the fifth region was discretized into 4 zone radially. Increasing the discretization in the axial or radial directions did not significantly alter the results. Thus, this level of discretization was considered adequate for resolving the shock waves that developed.

The simulation was run in Eulerian mode axially and in ALE mode radially. This was done to prevent the computational grid from curling back on itself. The SESAME equation of state table was used for simulating the silver equation of state and transport parameters. The SESAME tables were also used for the DT simulation except for the conductivity, in which a Spitzer-analytical model was used.

Figure 2 shows the contours of ion density and ion temperature during the run-in-phase at 80 ns and 100 ns. The outer liner has moved from 0.3cm to 0.25cm by 80 ns and 0.2cm by 100 ns. The generation of shock waves is visible in these contours both in the liner and target plasmas. When the implosion reaches at 80 ns, a sharp layer of Ag plasma separates from the main liner and stagnates at the outer surface of target plasma at 0.19cm. By 100 ns this layer at 0.12 cm grows thicker and hotter and becomes a second piston that compresses the target plasma in a very stable manner. The shock waves also propagate with a much faster speed in the target plasma and as it propagates radially inward it pre-heats the target plasma. A shock front in the target plasma reaches to 0.05 cm at 100 ns and the ion temperature due to shock heating rises to 34 eV.

Figure 3 shows contours of ion density and ion temperature near peak implosion (120 ns) and at the peak implosion (126.4 n). At 120 ns the liner plasma implodes to 0.06 cm showing the sign of RT instability, whereas the target plasma extends from axis to 0.03 cm remains uniform. The density of target plasma reaches above the solid density and temperature reaches above 100 eV. On the other hand the liner plasma remains relatively cold at an average temperature of 40 eV. The peak implosion time is 126.4 ns and is shown on the expanded radial scale. At this time the liner plasma compresses to the final radius of 0.008 cm and target plasma compresses to the final radius of 0.003 cm. Thus the compression ratio for the liner plasma is about 37 and for the target plasma is 72. The density of target plasma reaches almost 9×10^{24} cm^{-3} and average temperature reaches approximately to 4 keV.

Figure 4 shows the profiles of ion density, ion temperature, current density and magnetic field versus radius on expanded scale at 120 ns. The liner plasma peaks at 0.07 cm whereas the target plasma is confined within the radius of 0.025 cm. The maximum ion temperature of target plasma reaches 115 eV with an average density of 1.0×10^{23} cm^{-3}. A sharp layer of Ag plasma carrying the current density of 2×10^{13} A/m^2 that reverses in the negative direction and reverses again in the liner plasma. The trapped magnetic flux between this shocked layer and liner plasma compresses as the implosion proceeds. At this time it reaches to the value of 24 MG where as the magnetic field at the outer surface of liner plasma is about 43 MG.

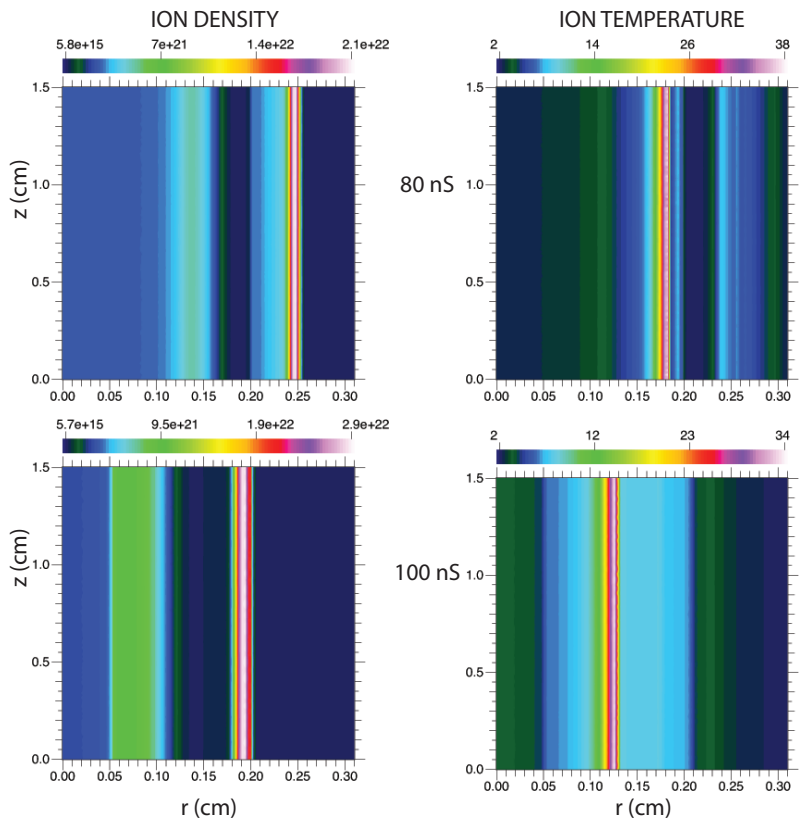

Fig. 2. Contours of ion density (left) and temperature (right) during at 80 ns (run-in) and 100 ns (peak compression).

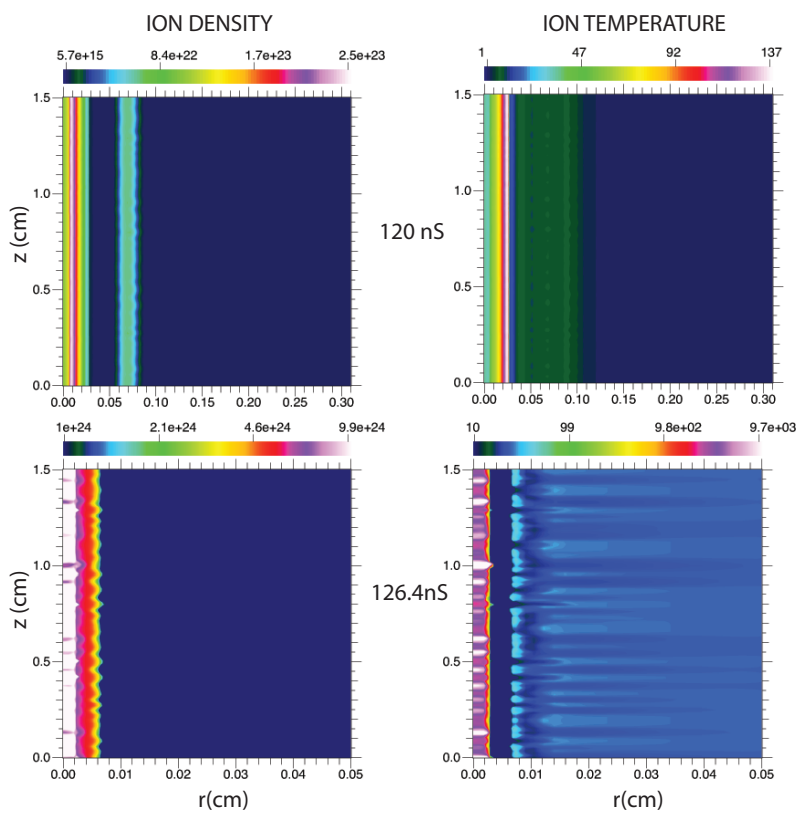

Fig. 3. Contours of ion density (left) and temperature (right) during at 120 ns (run-in) and 126.4 ns (peak compression).

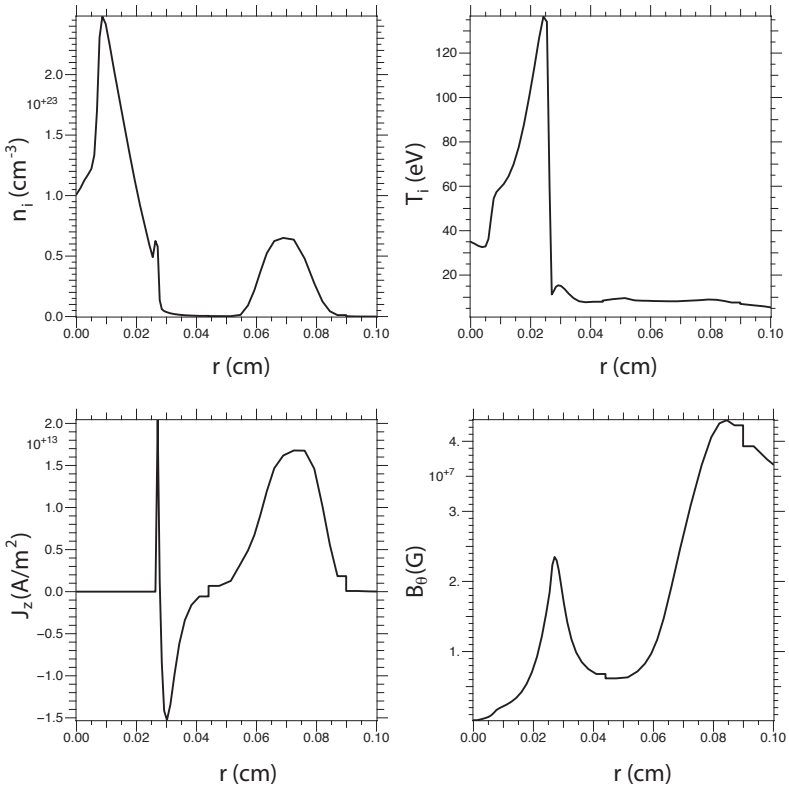

Fig. 4. Line-outs at 120 ns of ion density, ion temperature, current density and azimuthal magnetic field at versus r on an expanded scale.

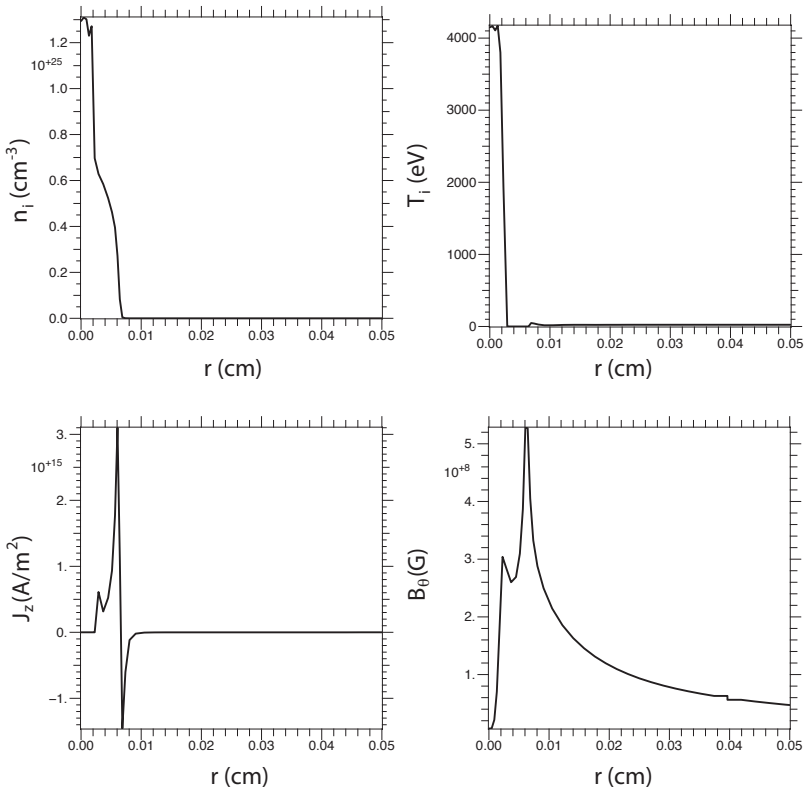

Fig. 5. Line-outs at 126.4 ns of ion density, ion temperature, current density and azimuthal magnetic field at versus r on an expanded scale.

Figure 5 shows the profiles of ion density, ion temperature, current density and magnetic field versus radius on further expanded scale at the peak implosion time of 126.4 ns. The liner plasma peaks at 0.007 cm whereas the target plasma is confined within the radius of 0.003 cm. The maximum ion temperature of target plasma reaches 4 keV with an average density of 1.3×10^{25} cm^{-3}. A sharp layer of Ag plasma carrying the current density of 6×10^{14} A/m^2 that reverses and reverses again in the liner plasma. The trapped magnetic flux between this shocked layer and liner plasma compresses and at this peak implosion time it reaches to a value of 300 MG where as the the magnetic field at the outer surface of liner plasma is about 500 MG. The α particles produced by fusion reaction can easily be trapped radially due to such large magnetic field and triggers the ignition. For the next one nano second time span the major heating is due to the deposition of these α particles and plasma heats up and explodes violently.

Figure 6 shows the plots of fusion energy E_f, kinetic energy E_k and ion energy E_i as function of time. The energy from the circuit is converted to the kinetic energy which reaches to a maximum level of 1 MJ. The plasma is heated initially due to ohmic heating, than shock heating with a simultaneous adiabatic compressional heating. But once the fusion reaction starts the α-particle heating raises the ion energy up to 10 MJ. The total fusion energy including α-particle and neutrons produced in this run is above 200 MJ thus providing a real gain of more than 200 if we consider kinetic energy as an input, but 9 if we consider the total stored energy of 22 MJ.

III. CONCLUSIONS

Shock waves play crucial role in mitigating RT instability leading to an effective energy coupling from the high power MJ class current generators. This has been discussed in a recent paper [4]. In this paper, we have presented a similar mechanism for producing high energy density plasma that leads to ignition with a modest compression ratio of 37 for Sandia ZR generator. 2D simulations using the MACH2 code for realistic load and currently available machine parameters. A compression ratio of 30 to 40 has been achieved routinely in various Z-pinch facilities, however this level of compression ratio alone cannot heat the plasma to keV range uniformly. The kind of staging described above where a high Z-liner plasma compresses a low Z-target plasma may lead to a significantly high temperature even with this level of modest compression ratio.

The formation and study of such plasma are of great interest for fusion. Shock waves play a crucial role in the dynamics, energy coupling, and the formation of a stable high energy density uniform plasma. The shock waves are predominantly magnetosonic in nature and can transport currents and magnetic fields across the radial profile much faster than the classical diffusion time scales.

The role of shock waves becomes significant when a dense high-Z liner plasma implodes on a low density hydrogen type target plasma. The propagation of shock waves in the liner plasma stagnates their energy at the interface of liner and target plasmas, which creates a separate stable implosion of the target plasma leading to a high-energy density plasma. As the implosion proceeds, the current builds up in this separated layer, which develops due to shock stagnation and the azimuthal magnetic flux keeps compressing, reaching to large values above 300 MG at the peak implosion. In the case of a DT target, the charged particles produced in the thermonuclear fusion reaction can easily be trapped by this isolating magnetic field and deposit their energy into the target plasma. This leads to ignition with a fusion energy output of 200 MJ with net gain of 100 if the input is used as kinetic energy and 9 if we use the total stored energy in the capacitor bank. A neutron yield of, $Y_n = 7 \times 10^{19}$, is predicted in this simulation run.

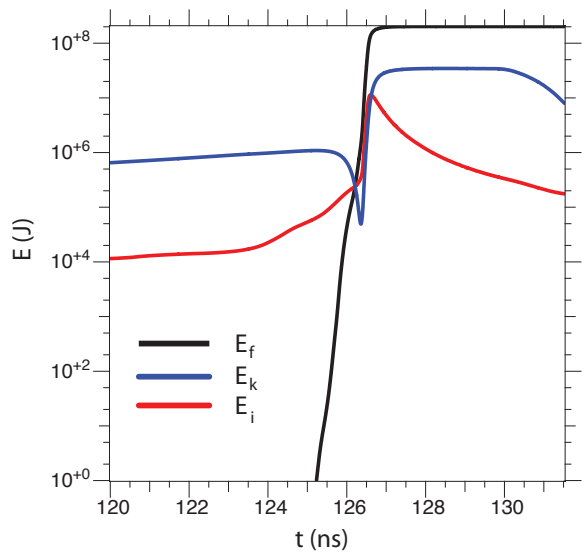

Fig. 6. Total fusion energy E_f, kinetic energy E_k and ion energy E_i is plotted versus time

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